

Design of an Instrumented Robot Data-Collection Cell for Physical AI

ME 423-X2

Individual Concept and Subsystem Analysis

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"I pledge my honor that I have abided by the Stevens Honor System."

Subsystem

This report discusses the static camera field of view (FoV) and testing rig of the main insertion data-collection system of our team. The camera captures the insertion and turning process, while the testing rig holds the lock in place during repeated trials.

The static camera we propose is an Intel RealSense D435i, with RGB and depth imaging. The GoPro HERO8 Black attaches to the handheld device separately. The static camera provides an alternate view when the handheld camera is blocked by the user's hand or gripper.

Instead of a door for each setup, the test rig will have a lock module that can be replaced. This matches the team's design documentation that includes interchangeable lock cylinders, adjustable fixtures, and rigid mounting hardware.

Design Metrics

The design metrics are based on our team's Phase 1 requirements for visual data collection, repeatability, interchangeable locks, safety, and cost.

Table 1. Design Metrics and Target Specifications

Customer Need	Design Metric	Target Value
N3, N6	Visible workspace	$\geq 0.40 \times 0.35$ m
N3	Image resolution at lock	≥ 2.0 pixels/mm
N10, N22, N28	Camera and lock positioning	Fixed camera, interchangeable lock
N6	Complete episode capture rate	$\geq 95\%$
N31	Tipping safety factor	≥ 1.5
N30	Material cost	Within project hardware budget

The 95% recording target comes directly from the Phase 1 report. The workspace, image resolution, and tipping safety-factor targets are proposed criteria for this analysis rather than finalized Phase 1 numerical specifications.

Detailed Design

The proposed static camera is an Intel RealSense D435i positioned 0.55 m from the lock. According to its manufacturer's datasheet, the RGB camera has a horizontal FoV of 69.4°, vertical FoV of 42.5°, and resolution of 1920 × 1080 pixels.

The D435i also supports depth sensing, with a nominal depth FoV of approximately 87° × 58°. The camera will be attached to a rigid mount or tripod to prevent movement after calibration, as identified in the team's design documentation.

The testing rig consists of a rigid stand with a replaceable lock module. The team's design documentation includes a door panel, lock cylinder, adjustable lock-height fixture, and adjustable door-orientation fixture. These components allow different lock configurations to be tested without rebuilding the entire setup.

The camera distance of 0.55 m is a proposed placement for the analysis. The final testing-rig dimensions and camera mounting location have not yet been selected.

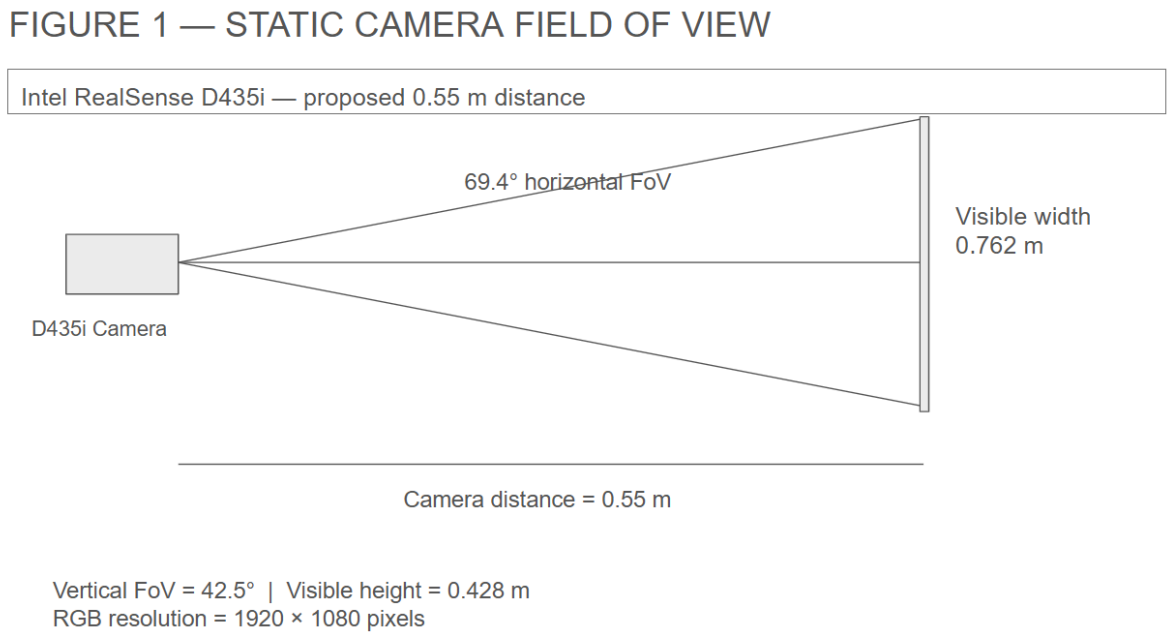


Figure 1. Static Camera Field of View

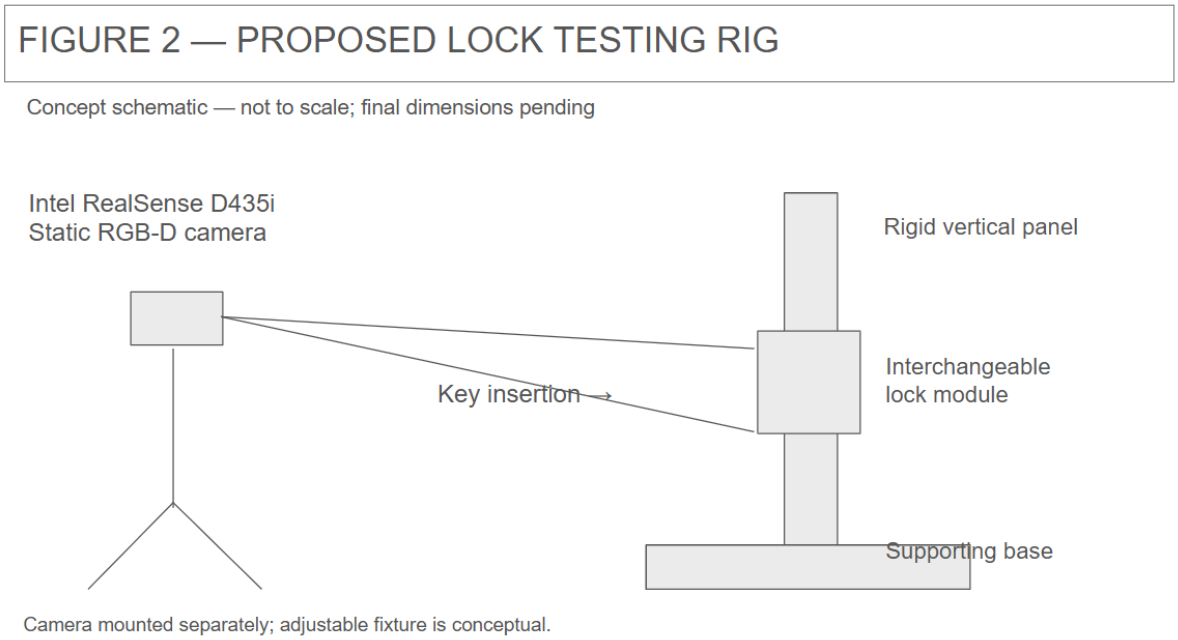


Figure 2. Lock Testing Rig

Analysis and Evaluation

The camera must capture the entire key insertion area. The visible width and height are calculated using the camera's field of view and distance from the lock.

Camera Field of View:

$$Visible\ Width\ =\ 2\ x\ Distance\ x\ tan(Horizontal\ FoV\ /\ 2)$$

$$W\ =\ 2\ x\ 0.55\ x\ tan(69.4°\ /\ 2)$$

$$W\ =\ 0.762\ m$$

$$Visible\ Height\ =\ 2\ x\ Distance\ x\ tan(Vertical\ FoV\ /\ 2)$$

$$H\ =\ 2\ x\ 0.55\ x\ tan(42.5°\ /\ 2)$$

$$H\ =\ 0.428m$$

The calculated visible area is approximately 0.762 × 0.428 m, exceeding the proposed workspace target of 0.40 × 0.35 m.

Image Resolution

$$Horizontal\ Resolution\ =\ 1920\ /\ 762\ =\ 2.52\ pixels/mm$$

$$Vertical\ Resolution\ =\ 1080\ /\ 428\ =\ 2.52\ pixels/mm$$

Both exceed the proposed target of 2.0 pixels/mm. These results assume the lock is centered in the camera's view and that the selected recording mode uses the full RGB field of view. Physical testing is still needed to check for obstructions from the user's hand and device.

Testing Rig Stability:

The testing rig must remain stable during key insertion. Tipping stability is evaluated by comparing the resisting and applied moments.

Resisting Moment:

$$M_R\ =\ m \times g \times (b\ /\ 2)$$

Applied Moment:

$$M_A\ =\ F \times h$$

Tipping Safety Factor:

$$SF\ =\ M_R\ /\ M_A$$

- m = testing-rig mass (kg)
- g = 9.81 m/s²
- b = base width in the tipping direction (m)
- F = horizontal applied force (N)
- h = height of applied force (m)

Combining these equations gives the minimum mass required for a proposed safety factor of 1.5:

$$m_min\ =\ (1.5 \times F \times h)\ /\ (9.81 \times b\ /\ 2)$$

This equation can be used to determine the required rig mass after the base dimensions and expected forces are selected. The team's documents do not yet provide a finalized testing-rig mass, base width, or expected insertion force. Therefore, a numerical tipping safety factor cannot currently be confirmed.

Manufacturing Cost:

The team's Phase 1 report identifies a $700 Senior Design budget and an estimated additional $2,000–$5,000 in laboratory funding [1]. The team also has access to equipment supplied by the laboratory, reducing the need to purchase certain components.

Table 2. Proposed Testing Rig Components

Component	Status
Door panel or rigid test stand	Proposed
Replaceable lock cylinder	Identified in team design
Adjustable lock fixture	Proposed
Camera tripod or rigid mount	Identified in team design
Mounting hardware	Pending final selection

Since the materials and final dimensions have not been selected, the testing-rig cost cannot yet be determined. The previous $55 estimate has been removed because it was not supported by a bill of materials or vendor quotations.

Table 3. Design Evaluation

Metric	Result	Target	Status
Visible workspace	0.762 × 0.428 m	≥ 0.40 × 0.35 m	Pass*
Image resolution	2.52 pixels/mm	≥ 2.0 pixels/mm	Pass*
Complete episode capture rate	Not yet measured	≥ 95%	Pending
Tipping safety factor	Not yet determined	≥ 1.5	Pending
Material cost	Not yet determined	Within project budget	Pending

The camera meets the proposed theoretical coverage and resolution requirements. The remaining requirements require the final rig dimensions, loading conditions, component selection, or physical testing.

Broader Impacts

The testing rig was evaluated for safety using the tipping moment equations. The required rig mass depends on the base width, applied force, and height of the lock. These relationships can be used to design a stable fixture once the remaining dimensions and forces are established.

The Phase 1 report identifies a $700 Senior Design budget with additional laboratory support. Using existing laboratory equipment and interchangeable lock fixtures may reduce additional purchases, although the actual savings cannot be quantified until the bill of materials is complete.

Overall, the proposed static camera meets the preliminary analytical requirements for field of view and image resolution. The testing rig is not yet validated, since its final dimensions, applied forces, and material costs are still being determined. These results provide a starting point for the team's Phase 2 design.

References

[1] Group C4, Phase 1 Report: Design of an Instrumented Robot Data-Collection Cell for Physical AI, 2026.

[2] Group C4, Handheld Device Design Doc, 2026.

[3] Intel, Intel RealSense D400 Series Product Family Datasheet, RGB Sensor Specifications.

[4] Intel, RealSense Depth Camera D435i Specifications.